Efficient quantum simulation of random-coupling models by spectral amplification

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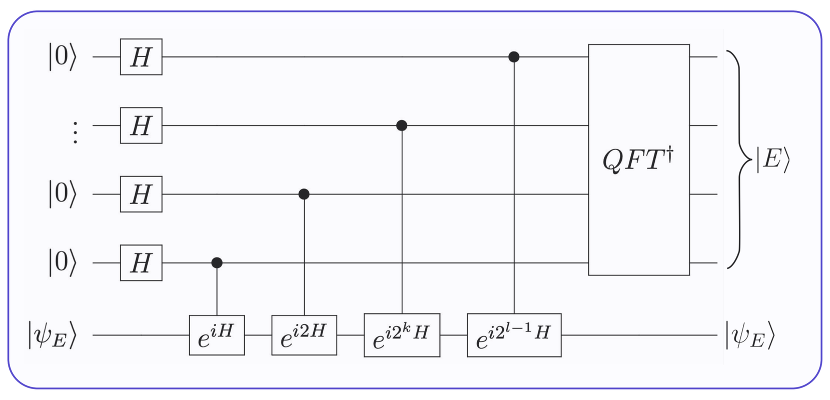
Spectral amplification accelerates quantum simulations

Hamiltonian simulation ↔ Energy measurement [1]



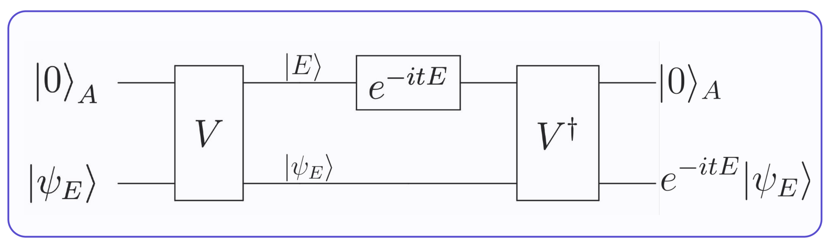
Energy measurement circuit by Quantum Phase Estimation

$$V|\psi_E\rangle|0\rangle_{\rm anc} \approx |\psi_E\rangle|E_{\rm est}\rangle_{\rm anc}$$

Longer time evolution for better precision

$$\delta E \sim 2^{-\ell} \sim 1/T$$

same gate complexity



Circuit V can be used for Hamiltonian simulation

$$t \sim 1/\delta E$$

Quadratic speedup by squaring the Hamiltonian $H = H_{\rm sqrt}^\dagger H_{\rm sqrt} + E_{\rm SA}$

- **Frustration-free Hamiltonians** [2]

$$H = \sum_X h_X \ (h_X \geq 0) \ \to \ H_{\rm sqrt} = \sum_X \sqrt{h_X} \otimes (|X\rangle\langle 0| + |0\rangle\langle X|)$$

- **Preprocessing by semidefinite programming (SDP)** [3]

$$\max_{G\succeq 0,\, E_{\rm SA}} \ E_{\rm SA} \quad \text{subject to} \quad H - E_{\rm SA} I = \sum_{a,b=1}^{L} G_{ab}\mathfrak{o}_a^\dagger \mathfrak{o}_b$$

$$H_{\rm sqrt} = \sum_m |m\rangle \otimes O_m \quad (H_{\rm sqrt}^\dagger H_{\rm sqrt} = \sum_m O_m^\dagger O_m)$$

Estimate of E to precision ϵ

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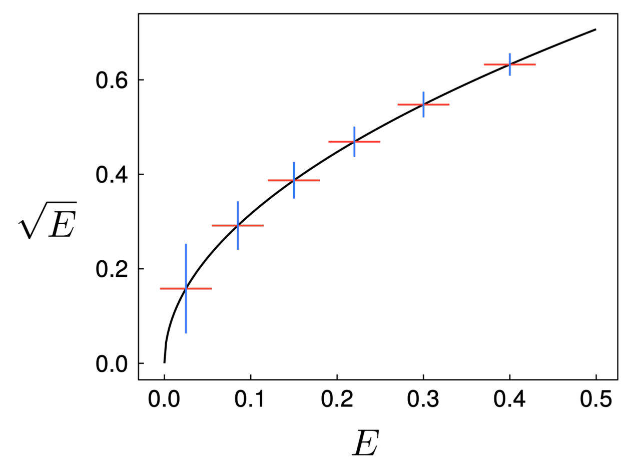
Estimate of $\sqrt{E}$ to precision $\mathcal{O}(\epsilon/\sqrt{E})$



Supersymmetric models: natural square roots enable spectral amplification

SUSY provides the square root operator

Q supercharge satisfying $\{Q, (-1)^F\} = 0$

$$H = Q^2 \quad (H_{\rm sqrt} = Q,\ E_{\rm SA} = 0)$$

e.g.,

$\mathcal{N} = 2$ Nicolai model $\quad \mathcal{Q} = \mathcal{Q} + \mathcal{Q}^\dagger,\ \mathcal{Q} = \sum_{k=1}^{L/2} c_{2k-1}c_{2k}^\dagger c_{2k+1}$

$\mathcal{N} = 1$ SUSY SYK model $\quad Q = \frac{i}{3!}\sum_{i,j,k} C_{ijk}\gamma_i\gamma_j\gamma_k$

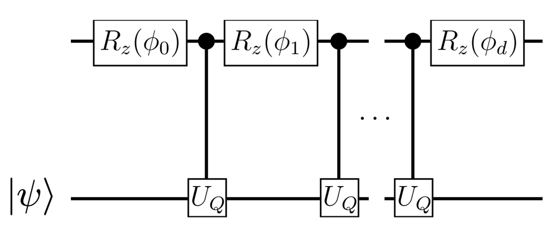
$\mathcal{N} = 2$ SUSY SYK model $\quad Q = \mathcal{Q} + \mathcal{Q}^\dagger,\ \mathcal{Q} = \frac{i}{3!}\sum_{a,b,c} C_{abc}c_a c_b c_c$

BPS states $\mathcal{Q}|\psi\rangle = 0$
connect to near-extremal supersymmetric black holes

Analogy with Topological Data Analysis
- nilpotent supercharge Q ↔ boundary map ∂
- BPS states ↔ homology classes

Hamiltonian simulation by QETU [4]

Quantum Eigenvalue Transformation of Unitary matrices



single-qubit rotations

oracle access to unitary $U_Q = \exp(-iQ/\lambda_{\rm sqrt})$

$$W = |0\rangle\langle 0| \otimes e^{-iQ/\lambda_{\rm sqrt}} + |1\rangle\langle 1| \otimes e^{+iQ/\lambda_{\rm sqrt}}$$

For SUSY models, since

$$\{Q, (-1)^F\} = 0 \implies (-1)^F U_Q (-1)^F = U_Q^\dagger$$

only two controlled-parity operations are required for

$$W = C(P)(I \otimes U_Q)C(P) \quad (C(P) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (-1)^F)$$

Gate complexity reduction for SUSY SYK

Target function (real part) $\quad f(x) = \cos(a \arcsin^2 x) \quad (a = t\lambda_{\rm sqrt}^2)$

Low-energy window $\in [0, \Gamma]$

Query complexity of QETU spectral amplification [5]

$$d = \mathcal{O}\left(t\sqrt{\Gamma\lambda_{\rm sqrt}^2} + \sqrt{\lambda_{\rm sqrt}^2\Gamma^{-1}}\log\frac{1}{\epsilon}\right)$$

Total gate complexity for SUSY SYK:

Trotterization to precision $\mathcal{O}(\epsilon/d) \times$ Query complexity d

$$G_{\rm 1st} = \widetilde{\mathcal{O}}\left[\frac{N^6}{\epsilon}\left(t\sqrt{\Gamma} + \frac{1}{\sqrt{\Gamma}}\log\frac{1}{\epsilon}\right)^2\right],$$

$$G_{\rm high} = N^{4+o(1)}\left(t\sqrt{\Gamma} + \frac{1}{\sqrt{\Gamma}}\log\frac{1}{\epsilon}\right)^{1+o(1)}\epsilon^{-o(1)}.$$

cf. naive Trotterization $\quad G_{\rm 1st} = \widetilde{\mathcal{O}}\left(N^{13/2}t^2/\epsilon\right),\quad G_{\rm high} = N^{9/2+o(1)}t^{1+o(1)}\epsilon^{-o(1)}.$

Sachdev-Ye-Kitaev (SYK) model: SDP-free sum-of-squares construction

SOS representation of SYK Hamiltonian

$$H_{\rm SYK} = \sum_{1\leq i<j<k<l\leq N} J_{ijkl}\gamma_i\gamma_j\gamma_k\gamma_l \qquad \mathbb{E}\left[J_{ijkl}^2\right] = \frac{6J^2}{N^3}$$

Quadratic Majorana blocks $B_m = i\sum_{1\leq i<j\leq N} A_{ij}^{(m)}\gamma_i\gamma_j$

$$A_{ij}^{(m)} \sim \mathcal{N}\left(0, \frac{J}{\sqrt{2MN^3}}\right)$$

Sum-of-Squares rep. without SDP optimization

$$H = \sum_{m=1}^{M} B_m^2 + E_{\rm SA} \quad (E_{\rm SA} = -\sum_m\sum_{i<j}(A_{ij}^{(m)})^2)$$

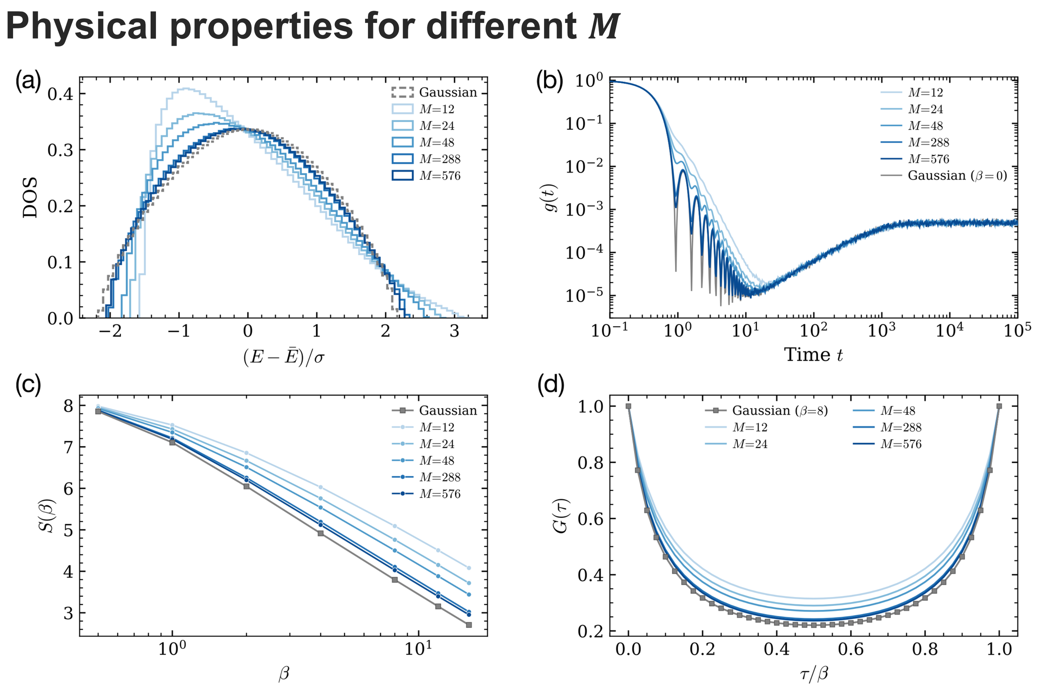
$$J_{ijkl} = -2\sum_{m=1}^{M}\left(A_{ij}^{(m)}A_{kl}^{(m)} - A_{ik}^{(m)}A_{jl}^{(m)} + A_{il}^{(m)}A_{jk}^{(m)}\right)$$

Interpolation between different regimes [6]

$M = \Theta(N)$
- Low-rank (class IV) SYK
- maximally chaotic
- tunable IR behavior (scaling dim.)

$M = \Theta(N^\alpha) \quad (\alpha > 1)$
- Dense SYK at large-N (same SD equations)

Physical properties for different M



Random generation of blocks

$$B_m \stackrel{\rm d}{=} U_m\left(\sum_{j=1}^{N-1}\tau_j^{(m)}i\gamma_j\gamma_{j+1}\right)U_m^\dagger, \quad R_m \sim {\rm Haar}({\rm SO}(N)),\ \ |\tau_j^{(m)}|/\sqrt{v} \sim \chi_{N-j}$$

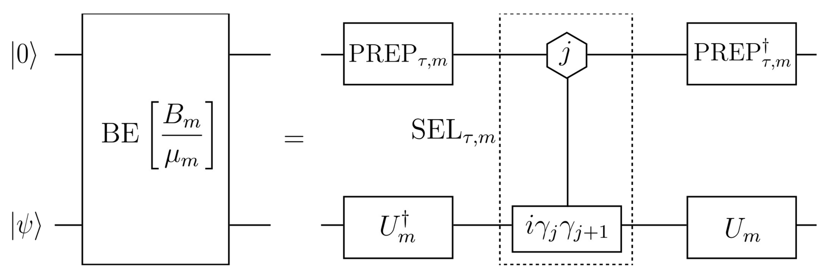
$$U_m\gamma_i U_m^\dagger = \sum_{j=1}^{N}(R_m)_{ij}\gamma_j$$

Efficient Block Encoding (BE)

BE of single block B_m

$$\mathrm{SEL}_{\tau,m} = \sum_{j=1}^{N-1}|j\rangle\langle j| \otimes \mathrm{sgn}(\tau_j^{(m)})i\gamma_j\gamma_{j+1}$$

$$\mathrm{PREP}_{\tau,m}|0\rangle = \frac{1}{\sqrt{\mu_m}}\sum_{j=1}^{N-1}\sqrt{|\tau_j^{(m)}|}\,|j\rangle \qquad \mu_m = \sum_{j=1}^{N-1}|\tau_j^{(m)}|$$



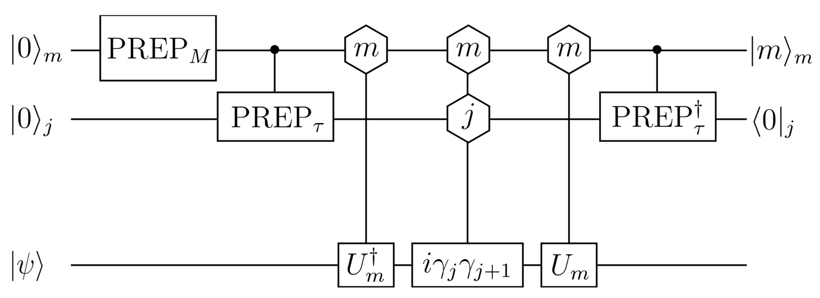
BE of square root operator $H_{\rm sqrt}$

$$\mathrm{PREP}_M|0\rangle_m = \frac{1}{\lambda_{\rm sqrt}}\sum_{m=1}^{M}\mu_m|m\rangle_m \qquad \lambda_{\rm sqrt}^2 = \sum_{m=1}^{M}\mu_m^2$$

$$\mathrm{SEL}_\tau = \sum_{m,j}|m,j\rangle\langle m,j| \otimes \mathrm{sgn}(\tau_j^{(m)})i\gamma_j\gamma_{j+1}$$

$$\mathrm{PREP}_\tau|m\rangle_m|0\rangle_j = |m\rangle_m\frac{1}{\sqrt{\mu_m}}\sum_{j=1}^{N-1}\sqrt{|\tau_j^{(m)}|}\,|j\rangle_j$$

$$\mathcal{U} = \sum_{m=1}^{M}|m\rangle\langle m| \otimes U_m$$



$$\mathcal{V} := \mathrm{PREP}_\tau^\dagger\,\mathcal{U}\,\mathrm{SEL}_\tau\,\mathcal{U}^\dagger\,\mathrm{PREP}_\tau\,\mathrm{PREP}_M$$

$$(I_m \otimes \langle 0|_j \otimes I)\,\mathcal{V}\,(|0\rangle_m \otimes |0\rangle_j \otimes I) = \frac{H_{\rm sqrt}}{\lambda_{\rm sqrt}}$$

Implementation of $\mathcal{U}$

$$U_m = G_K(\theta_K^{(m)})\cdots G_2(\theta_2^{(m)})G_1(\theta_1^{(m)}), \quad K = \frac{N(N-1)}{2}$$

$$|m,k,0\rangle|\psi\rangle \xrightarrow{O_\theta} |m,k,\theta_k^{(m)}\rangle|\psi\rangle \xrightarrow{G_k(\theta_k^{(m)})} |m,k,\theta_k^{(m)}\rangle G_k(\theta_k^{(m)})|\psi\rangle \xrightarrow{O_\theta^\dagger} |m,k,0\rangle G_k(\theta_k^{(m)})|\psi\rangle.$$

On-the-fly implementation of each rotation

Overall Complexity

Normalization $\quad \lambda_{\rm sqrt}^2 = \Theta(J\sqrt{M}N^{3/2})$

SOS lower bound $\quad |E_{\rm SA}| = \Theta(J\sqrt{MN})$

Assume $\quad \Gamma = \Theta(|E_{\rm SA}|)$,

Query complexity

$$d = \mathcal{O}\left(t\lambda_{\rm sqrt}\sqrt{\Gamma} + \frac{\lambda_{\rm sqrt}}{\sqrt{\Gamma}}\log\frac{1}{\epsilon}\right) = \mathcal{O}\left(JtN\sqrt{M} + \sqrt{N}\log\frac{1}{\epsilon}\right)$$

Gate complexity

$$G = \widetilde{\mathcal{O}}\left(JtN^3\sqrt{M} + N^{5/2}\log\frac{1}{\epsilon}\right)$$

QROM-free implementation of rotation unitary U_m reduces gate complexity of the oracle [7] $\mathcal{O}(MN^2) \to \widetilde{\mathcal{O}}(N^2)$

Summary and Outlook

SUSY models
- Supercharge as natural square root operator $\quad H = Q^2$
- QETU enables BE-free spectral amplification with one ancilla

Dense SYK models
- Random quadratic blocks provide SOS rep. without SDP

$$H = \sum_{m=1}^{M} B_m^2 + E_{\rm SA}$$

- Coefficients are generated on the fly, eliminating QROM loading

Beyond SYK models
- Application to other random coupling models, e.g., spin-SYK-type models, random-exchange spin glasses

$$H = \sum_{\{i_q\}}\sum_{\alpha_q = x,y,z} J_{\{i_q\}}^{\{\alpha_q\}}\sigma_1^{\alpha_1}\cdots\sigma_{i_q}^{\alpha_q} \qquad H = \sum_{i,j} J_{ij}\boldsymbol{S}_i \cdot \boldsymbol{S}_j$$

Selected References

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[7] King et al., Phys. Rev. Lett. 136, 110601 (2026).